

EMPIRICAL ASSET PRICING

Doctoral Seminar

FNAN 883

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Course Description

This course is intended for Ph.D students. We will review the state of the empirical asset pricing literature devoted to understanding cross-sectional differences in asset returns. Both methodologies and empirical evidence will be surveyed.
The following topics will be covered:

- Single and multi factor asset pricing models: theoretical and empirical aspects.
- Estimating and evaluating asset pricing models.
- Time-series versus cross-section methods in empirical asset pricing.
- Empirical methods for documenting 'anomalies', i.e. deviations from predictions of asset pricing models.
- Time-series anomalies: the equity premium, risk free rate, and excess volatility puzzles.
- Cross-sectional anomalies: size, value, price momentum, earnings momentum, dispersion, credit risk, idiosyncratic volatility effects, and others.

- Explanations for asset pricing anomalies: risk-based, estimation-based, market frictions, behavioral.

Course materials

Cochrane, John H., 2001, *Asset Pricing*. Princeton University Press, Princeton, New Jersey, 2005

Campbell, John. Y., Andrew W. Lo, and A. Craig MacKinlay, *The Econometrics of Financial Markets*, Princeton University Press, 1997

A great survey of empirical asset pricing

Goyal, Amit, 2012, "Empirical Cross-Sectional Asset Pricing: A Survey." *Financial Markets and Portfolio Management*, 26 (1), 3–38.

1 Asset pricing models

1.1 Overview

- We will address theoretical and empirical aspects of the time-series and cross-sectional properties of expected stock returns.
- Statistically, an expected asset return can be formulated as

$$\mathbb{E}(r_{i,t}) = \alpha_i + \beta_i' \mathbb{E}(f_t) \tag{1}$$

  where f denotes a set of K factor-mimicking portfolios, β_i is a K vector of factor loadings, and α_i reflects the expected return component unexplained by factors (a.k.a. model mispricing).
- For example, the market model is a statistical model with f being represented by the market portfolio.
- An asset pricing model intends to identify economic [(I)CAPM] or statistical [APT] common factors that *eliminate* α.
- In the absence of alpha, expected return differential across assets is triggered by factor loadings only.

- The presence of asset mispricing could give rise to additional cross-sectional differences in expected returns.
- The empirical evidence shows that
  - α **is there**
  - α **varies cross-sectionally** with firm-level variables such as size, the book-to-market ratio, past returns, and other firm characteristics – these cross-sectional patterns in average stock returns are classified as **"anomalies"**, since asset pricing models cannot explain them.
  - α **varies over time** with both business conditions (proxied by variables such as the default spread and the term spread), firm-level variables, and investor sentiment (Baker and Wurgler, 2006)

1.2 CAPM

- The CAPM of Sharpe (1964) and Lintner (1965) originated the literature on asset pricing models.
- The CAPM is a general equilibrium model in a single-period economy.

The model imposes an economic restriction on the statistical structure in (1) and it says that

$$\mathbb{E}(r_{i,t}) = \text{cov}(r_{i,t}, r_{m,t}) \frac{\mathbb{E}(r_{m,t})}{\text{var}(r_{m,t})}, \tag{2}$$

$$= \beta_{i,m} \mathbb{E}(r_{m,t}), \tag{3}$$

where $r_{m,t}$ is excess return on the market portfolio at time t.

- The CAPM measures an **asset risk** as the **covariance** of the asset return with the market portfolio return.
- The higher the covariance the less desirable the asset is and thus the expected return is higher.

- The **market price of risk**, common to all assets, is determined in equilibrium by the **risk aversion of investors** and the **volatility of the market**.
- For practical implications such as
  - estimating the cost of equity capital
  - testing the CAPM
  - evaluating the performance of active management such as mutual funds, hedge funds, and investment newsletters

  the market portfolio is commonly proxied by the value-weighted NYSE portfolio, S&P500 index, or CRSP index (AMEX, NYSE, and NASDAQ combined).
- Overall, the CAPM is nice, simple, and intuitive.
- But there are just too many empirical and theoretical drawbacks.

1.2.1 Empirical drawbacks

A. Existence of Anomalies – the CAPM is at odds with anomalous patterns in the cross section of stock returns - both corporate and asset pricing anomalies. **Firm-level variables** predict cross-sectional differences in returns. They are called anomalies because **beta cannot explain these differences.**

Prominent **asset pricing anomalies** include the size, book-to-market, past returns (momentum, reversal), earnings momentum, dispersion, accruals, credit risk, idiosyncratic volatility, asset growth, and investment anomalies.

The size and value effects

Basu (1977), Banz (1981), and Fama and French (1992) [FF] suggest that the CAPM cannot explain the effect of size, the dividend yield, the earnings yield, and the book-to-market on average returns.

- **Size effect**: small stocks have higher average returns than large stocks.
- **Value effect**: value stocks have higher average returns than growth stocks.

Value (growth) firms have high (low) E/P, B/P, D/P, or CF/P.
Why 'value'? Because physical assets can be purchased at low prices.
Why 'growth'? Because high price relative to fundamentals reflects capitalized growth opportunities.

Past return anomalies

- **Short-term reversal** – Lehmann (1990) and Jegadeesh (1990) show that 'contrarian strategies' that exploit the short-run return reversals in individual stocks generate abnormal returns of about 1.7% per week and 2.5% per month. *Contrarian strategies* involve buying stocks that have performed the worst and selling stocks that have performed the best over the past.
- **Momentum (medium-term momentum)** – Jegadeesh and Titman (1993) and a great body of subsequent work uncover abnormal *stock* returns to 'momentum strategies' with investment horizons of 3, 6, 9, and 12 months.
  In particular, stocks that have outperformed over the past 3, 6, 9, and 12 months tend to outperform over the subsequent 3, 6, 9, and 12 months. *Momentum strategies* involve buying stocks that have performed the best and selling stocks that have performed the worst over the past. Mo-

mentum is the most profitable and robust asset-pricing anomaly. Momentum is also present in *U.S. corporate bonds* (see Jostova, Nikolova, et al., 2013). It is also profitable in *international equities* (see Rouwenhorst, 1998), as well as in *currencies, commodities, and real estate* (e.g. Asness, Moskowitz, and Pedersen, 2013).

– **Long-term reversal** – De Bondt and Thaler (1985) and De Bondt and Thaler (1987) document long run reversal in U.S. stock returns over 2 to 5 years.

Accounting based anomalies

– **Earnings momentum** – Ball and Brown (1968) document the post-earnings-announcement drift (PEAD or earnings momentum). In particular, firms reporting unexpectedly high earnings subsequently outperform firms reporting unexpectedly low earnings over a 3, 6, and 9 month holding period.

– **Accruals** – Sloan (1996) shows that firms with relatively high (low) levels of accruals to total assets experience negative (positive) future abnormal returns. In particular, taking long (short) position in firms

with the lowest (highest) accruals yields about 10% per year over a one-year holding period.

- **Capital investments** – Titman, Wei, and Xie (2004) document that firms in high capital investments have lower stock returns than firms with low capital investments over a one year holding period. Capital investment is measure by the ratio of capital expenditure to property, plant, and equipment.
- **Asset growth** – Cooper, Gulen, and Schill (2008) find that firms with high asset growth have lower stock returns than firms with low asset growth over the subsequent year. Asset growth is measure by the change in total assets.

Credit conditions

- **Credit risk level** – Dichev (1998), Campbell, Hilscher, and Szilagyi (2008), and Avramov, Chordia, Jostova, et al. (2009a) demonstrate a *negative* cross-sectional correlation between credit risk and future stock returns.
- **Credit risk changes** – Dichev and Piotroski (2001) document abnormal stock price declines following credit rating downgrades. Interestingly, stock prices do not advance following rating upgrades.

Other anomalies

- **Dispersion** – Diether, Malloy, and Scherbina (2002) suggest that firms with high dispersion in analysts' earnings forecasts have lower stock returns than firms with low dispersion for holding periods of 1-3 months.
- **Idiosyncratic volatility [IV]** – Ang, Hodrick, Xing, and Zhang (2006) and Ang, Hodrick, Xing, and Zhang (2009) show negative cross section relation between idiosyncratic volatility and average return in both U.S. and global markets.

Counter intuitive relations: The dispersion, credit risk, and IV effects apparently violate the risk-return tradeoff. High risk (though not necessarily

systematic) is related to lower stock returns.

Are anomalies pervasive?

- Lo and MacKinlay (1990) claim that the size effect may very well be the result of unconscious, exhaustive search for a portfolio formation creating with the aim of rejecting the CAPM.
- Hou, Xue, and Zhang (2020): 65% of 452 anomalies fail to replicate.
- McLean and Pontiff (2016): academia destroys anomalies.
- Lettau and Ludvigson (2001a) and Lettau and Ludvigson (2001b) suggest that the size and value effects are consistent with a conditional version of the (C)CAPM with the consumption to wealth ratio (cay) being the conditioning variable.
- Jostova and Philipov (2005) show that a model with stochastic betas explains the size and value effects.
- Avramov, Chordia, and Goyal (2006): the short-term reversal strategies yields profits that do not survive transactions costs.
- Avramov, Chordia, Jostova, and Philipov (2007), Avramov, Chordia, Jostova, and Philipov (2009a), Avramov, Chordia, Jostova, and Philipov (2009b), and Avramov, Chordia, Jostova, and Philipov (2013) show that

the momentum, dispersion, and credit risk effects concentrate in a very small portion of high credit risk stocks, which are highly illiquid.

– Two papers analyze asset pricing anomalies in a unified framework: Fama and French (2008) and Avramov, Chordia, Jostova, et al. (2013).

– Analysts cause anomalies: Grinblatt, Jostova, and Philipov (2025).

B. Multifactor extensions of the CAPM such as Chen, Richard Roll, and Ross (1986), Fama and French (1993), and Pástor and Stambaugh (2003) associate expected returns with multiple risk factors.

- Multifactor models explain better the cross section of stock returns: Fama and French (2015), Hou, Xue, and Zhang (2015).
- However,
  - Multifactor models need not perform better out-of-sample.
  - The economic appeal of the additional factors is sometimes questionable.

C. Evidence from mutual and hedge funds: some mutual and hedge funds seem to **outperform** benchmark indexes, even after controlling for market risk. However, multifactor extensions explain most fund persistence (Carhart (1997) focuses on mutual funds and Fung and Hsieh (2001) and Fung and Hsieh (2004) on hedge funds).

D. R. Roll (1977) claims that the CAPM is untestable since the market return is unobservable and it cannot be measured accurately. Using proxy for the market factor could deliver unreliable inference
$\Rightarrow$ **the CAPM is not testable.**

1.2.2 Theoretical drawbacks of the CAPM

A. The CAPM assumes that the average investor cares only about the performance of his investment portfolio.

- But eventual wealth could come from both investment and **labor and entrepreneurial incomes** ⇒ an **additional factor** is therefore needed.
- The CAPM says that two stocks that are equally sensitive to market movements (i.e. have the same betas) must have the same expected return.
  - ∗ But if one stock performs better in recessions it would be more desirable for most investors who may lose their jobs or get lower salaries in recessions.

    ⇒ the investors will bid up the price of that stock, thus lowering its expected return.

    ⇒ **procyclical stocks should offer higher average returns than countercyclical stocks**, even if both stocks have the same market beta.

    ⇒ covariation with recessions should matter for expected returns.

- You may correctly argue that the market tends to go down in recessions,

  $\mapsto$ But recessions tend to be unusually severe or mild for a given level of market returns.

B. The CAPM applies to single-period (myopic) investors. Merton (1973) introduces the **Intertemporal CAPM [ICAPM]**, wherein the demand for risky assets is attributed in part to hedging against unfavorable shifts in the investment opportunity set.

- Merton points out that **an asset's risk should be measured by its covariance with the marginal utility of investors**, which need not be the same as the covariance with the market return.
- Merton shows that **multiple state variables that are sources of priced risk** are required to explain the cross section of expected returns.
- In intertemporal models, equilibrium expected returns on risky assets may differ from the riskless rate even when they have no systematic (market) risk.
- But Merton does not say which other state variables are priced.
- Most prominent in the class of intertemporal models is the **Consumption CAPM [CCAPM]** of Rubinstein (1976), Lucas (1978), Breeden (1979), and Grossman and Shiller (1988).
- The greatness of the CCAPM is identify an economically based state variable.

– The CCAPM is a single-state variable model – **real consumption growth is the single factor.**

1.3 The Consumption CAPM — CCAPM

- The CCAPM is also a theoretically motivated model.
- **The CCAPM replaces the market factor by consumption growth.**
- Let us derive a simple discrete time version of the CCAPM (see Cochrane, 2001).
- We use the power utility to describe investor preferences:

$$u(c_t) = \frac{1}{1-\gamma} c_t^{1-\gamma} \text{ for } \gamma \neq 1, \tag{4}$$

$$u(c_t) = \log(c_t) \text{ for } \gamma = 1. \tag{5}$$

  where:

  - c_t denotes consumption at time t
  - γ is the relative risk aversion parameter.

- The investor can freely buy or sell any amount of a security whose current price is p_t and next-period payoff is x_{t+1} ($x_{t+1} = p_{t+1} + d_{t+1}$).
- The investor must decide how to allocate wealth between consumption and saving $\Rightarrow$ **how much will she buy or sell?**

- To find the answer, consider a two-period investor whose income at time t is e_t, and let y be the amount of security she chooses to buy.

The investor's problem is

$$\max_y u(c_t) + \mathbb{E}_t \left[\rho u(c_{t+1})\right] \quad s.t. \tag{6}$$

$$c_t = e_t - p_t y, \tag{7}$$
$$c_{t+1} = e_{t+1} + x_{t+1} y, \tag{8}$$

ρ is the impatience parameter: the **'subjective discount factor'**.

- Substituting the constraints into the objective, and setting the derivative with respect to y equal to zero, we obtain **the first order condition** for an optimal consumption and portfolio choice,

$$p_t u'(c_t) = \mathbb{E}_t \left[\rho u'(c_{t+1}) x_{t+1}\right]. \tag{9}$$

- The left hand side in (9) reflects the loss in utility from consuming less as the investor buys an additional unit of the asset.

- The right hand side describes the expected increase in utility obtained from the extra payoff at $t+1$ attributed to this additional unit of the asset.
- A well-know representation for the first order condition is obtained by dividing both sides of (9) by $p_t u'(c_t)$. We obtain the fundamental discount factor view of asset pricing theories:

Discount Factor Asset-Pricing Representation:

$$1 = \mathbb{E}_t \left(\xi_{t+1} R_{t+1} \right), \tag{10}$$

where $\xi_{t+1} = \frac{\rho u'(c_{t+1})}{u'(c_t)}$ **is referred to as:**

the pricing kernel,

the marginal rate of substitution, or

the stochastic discount factor,

and

$R_{t+1} = \frac{x_{t+1}}{p_t}$ **denotes the gross return on the security.**

1.3.1 Several Insights about the First Order Condition

1. Observe from (10) that the gross risk free rate, the rate known at time t, which is uncorrelated with the discount factor, is given by

$$R^f_{t,t+1} = 1/\mathbb{E}_t(\xi_{t+1}). \tag{11}$$

2. When investor preferences are described by the power utility function, as in equation (4), the pricing kernel takes the form $\xi_{t+1} = \rho(c_{t+1}/c_t)^{-\gamma}$.

3. Assuming lognormal consumption growth one can show that the continuously compounded riskfree rate is (see Cochrane, 2001, chapter 1)

$$r^f_{t,t+1} = -\ln(\rho) - \ln\mathbb{E}_t\left[\exp(-\gamma\Delta \ln c_{t+1})\right],$$

$$r^f_{t,t+1} = -\ln(\rho) + \gamma\mathbb{E}_t(\Delta \ln c_{t+1}) - \frac{\gamma^2}{2}\sigma^2_t(\Delta \ln c_{t+1}). \tag{12}$$

4. To derive (12) we have used the useful relation that if x is normally distributed then

$$\mathbb{E}(e^{ax}) = e^{\mathbb{E}(ax)}e^{\frac{a^2}{2}\sigma^2(x)}. \tag{13}$$

5. A risky asset is one having nonzero correlation with the discount factor.

   From eq.(10),

$$
\begin{aligned}
1 &= \mathbb{E}_t\left(\xi_{t+1}R_{t+1}\right) \\
1 &= \mathbb{E}_t\left(\xi_{t+1}\right)\mathbb{E}_t\left(R_{t+1}\right) + \operatorname{cov}_t(R_{i,t+1},\xi_{t+1}) \\
\frac{1}{\mathbb{E}_t\left(\xi_{t+1}\right)} &= \mathbb{E}_t\left(R_{t+1}\right) + \frac{\operatorname{cov}_t(R_{i,t+1},\xi_{t+1})}{\mathbb{E}_t\left(\xi_{t+1}\right)} \\
R^f_{t,t+1} &= \mathbb{E}_t\left(R_{t+1}\right) + \frac{\operatorname{cov}_t(R_{i,t+1},\xi_{t+1})}{\mathbb{E}_t\left(\xi_{t+1}\right)} \\
\mathbb{E}_t\left(R_{t+1}\right) &= R^f_{t,t+1} - \frac{\operatorname{cov}_t(R_{i,t+1},\xi_{t+1})}{\mathbb{E}_t\left(\xi_{t+1}\right)} \\
\mathbb{E}_t\left(R_{t+1}\right) &= R^f_{t,t+1} + \left(\frac{\operatorname{cov}_t(R_{i,t+1},\xi_{t+1})}{\operatorname{var}_t(\xi_{t+1})}\right)\left(-\frac{\operatorname{var}_t(\xi_{t+1})}{\mathbb{E}_t(\xi_{t+1})}\right)
\end{aligned}
$$

Then, we obtain a **beta pricing model**

$$\mathbb{E}_t(r_{i,t+1}) = r^f_{t,t+1} + \underbrace{\left(\frac{\text{cov}_t(r_{i,t+1}, \xi_{t+1})}{\text{var}_t(\xi_{t+1})}\right)\left(-\frac{\text{var}_t(\xi_{t+1})}{\mathbb{E}_t(\xi_{t+1})}\right)}_{risk\ adjustment}. \tag{14}$$

$$\mathbb{E}(r_{i,t+1}) = r^f + \beta_{i,\xi}\lambda_\xi, \tag{15}$$

6. In words, expected excess return on each security should be proportional to the coefficient in the regression of that return on the discount factor, $\beta_{i,\xi}$.
7. The constant of proportionality, common to all assets, is the risk premium, λ_ξ.

8. Focusing on the power utility function and using a first order Taylor series expansion, we obtain (see Cochrane, 2001, ch.1)

The discrete time version of the consumption CAPM

$$\mathbb{E}(r_{i,t+1}) = r^f + \beta_{i,\Delta c}\lambda_{\Delta c}, \tag{16}$$

where

$$\beta_{i,\Delta c} = \frac{\text{cov}(r_{i,t+1}, \Delta c)}{\text{var}(\Delta c)}, \tag{17}$$

$$\lambda_{\Delta c} = \gamma \text{var}(\Delta c). \tag{18}$$

9. The relation is exact in continuous time.
10. The **asset's risk** is defined as the **covariance** between the asset return and consumption growth.
11. The **risk premium** is obtained as the product of the investor **risk aversion** and the **volatility of consumption growth**.

12. Notice from equation (16) that the expected return on an asset is larger the larger its covariance with consumption growth.
    Intuition: an asset doing badly in recessions (positive covariance) when the investor consumes little, is less desirable than an asset doing badly in expansions (negative covariance) when the investor feels wealthy and is consuming a lot. The former asset will sell for a lower price, thereby having a higher expected return.

1.3.2 Theoretically, the CCAPM appears preferable to the traditional CAPM

- It takes into account the dynamic nature of portfolio decisions
- It integrates the many forms of wealth beyond financial asset wealth
- Consumption should deliver the purest measure of good and bad times as investors consume less when their income prospects are low or if they think future returns will be bad.

Empirically, however, The CCAPM has disappointed as described below.

1.3.3 CCAPM: The Empirical Evidence

- Hansen and Singleton (1982) and Hansen and Singleton (1983) formulate a consumption-based model in which a representative agent has time-separable power utility of consumption.
  $\longmapsto$ they reject the model on U.S. data, finding that it cannot simultaneously explain the time-variation of interest rates and the cross-sectional variation of average stock and bond returns.
- Wheatley (1988) rejects the model based on international data.
- Mankiw and Shapiro (1986) show that the CCAPM performs no better, and in many respects even worse, than the CAPM. $\longmapsto$ They regress the average returns of the 464 NYSE stocks that were continuously traded from 1959 to 1982 on their market betas, on consumption growth betas, and on both betas. $\longmapsto$ They find that the market betas are more strongly and robustly associated with the cross section of average returns, and that market beta drives out consumption beta in multiple regressions.
- Breeden, Gibbons, and Litzenberger (1989) find comparable performance of the CAPM and a model that uses a **mimicking portfolio** for consumption growth as the single factor.

- Cochrane (1996) finds that the static CAPM substantially outperforms the consumption CAPM in pricing size portfolios.
- Academics and practitioners typically use portfolio-based models, such as the CAPM or the Fama and French extension to correct for risk.
- **Overall, the CCAPM has not been a great success.** As Cochrane (2001, p. 44) argues *"The consumption-based model is, in principle, a complete answer to all asst pricing questions, but works poorly in practice"*. **Consumption data is low frequency and too smooth.**
- Recently, Lettau and Ludvigson (2001b) and Amir and Bansal (2004) suggest that *Consumption Strikes Back!*. Indeed, Parker and Julliard (2005), Bansal, Dittmar, and Lundblad (2005), Hansen, Heaton, and Li (2008) suggest that if systematic consumption risk is measured **over long horizons** the CCAPM is able to explain the cross section dispersion in average stock returns.
- Moreover, **more volatile proxies for consumption are viable alternaties**. $\longmapsto$ For example, Savov (2011) shows that if consumption is proxied by *'garbage'*, a 'garbage-based' CCAPM works much better and evades the joint equity premium-risk free rate puzzle (see below).

- In the same spirit, Avramov, Chordia, Jostova, et al. (2012) show that a *World Credit Risk factor*, motivated by its connection to consumption risk, is able to price international equities and explains the higher equity returns of emerging market countries as well as the higher returns of high credit risk countries. Intuition: credit risk is countercyclical, since default losses are more likely in bad states of nature when consumption is low. Hence credit risk commands a risk premium. The credit risk premium is itself countercyclical – it is higher in recessions, which makes borrowing more costly, causing further declines in investment, future GDP growth and future consumption growth (see Gomes and Schmid, 2010). Hence credit risk is not only correlated with current consumption, it predicts future consumption growth.

1.4 The Consumption CAPM and the Equity Premium Puzzle

- From a cross-sectional perspective, the CCAPM fails if consumption beta is unable to explain why average returns differ across stocks, which is the case.
- At the aggregate level, the CCAPM leads to the so-called *equity premium puzzle* documented by Mehra and Prescott (1985) as well as the *risk free*

puzzle.

- To illustrate, let us manipulate the first order condition (10) (for notational clarity, I suppress the time dependence)

$$\begin{aligned} 1 &= E(\xi R), && (19)\\ &= E(\xi)E(R) + cov(\xi, R), && (20)\\ &= E(\xi)E(R) + \rho_{\xi,R}\sigma(\xi)\sigma(R). && (21) \end{aligned}$$

- Dividing both sides of (21) by $E(\xi)\sigma(R)$ leads to

$$\frac{E(R) - R^f}{\sigma(R)} = -\rho_{\xi,R}\frac{\sigma(\xi)}{E(\xi)}, \tag{22}$$

which implies that

$$\left|\frac{E(R) - R^f}{\sigma(R)}\right| \leq \frac{\sigma(\xi)}{E(\xi)} \left[= \sigma(\xi)R^f\right]. \tag{23}$$

- The left hand side in (23) is known as the **Sharpe ratio**.
- The highest Sharpe ratio is associated with portfolios lying on the mean-variance efficient frontier.

- Notice that **the slope of the frontier depends on the volatility of the discount factor** and **the risk free rate.**
- Under the CCAPM it follows that

$$\left|\frac{E(R^{mv}) - R^f}{\sigma(R^{mv})}\right| = \frac{\sigma\left[(c_{t+1}/c_t)^{-\gamma}\right]}{E\left[(c_{t+1}/c_t)^{-\gamma}\right]}. \tag{24}$$

- When log consumption growth is normally distributed the right hand side of (24) can be shown to be equal to (see Cochrane, 2001, p. 23) $\sqrt{e^{\gamma^2\sigma^2(\Delta \ln c_{t+1})}}$, which can be approximated by $\gamma\sigma(\Delta \ln c)$. Hence,

The slope of the mean variance frontier is determined by

$$\left|\frac{E(R^{mv}) - R^f}{\sigma(R^{mv})}\right| = \gamma\sigma(\Delta \ln c). \tag{25}$$

- In words, **the slope of the mean-variance efficient frontier is higher when the economy is riskier (consumption growth is more volatile)** or **when investors are more risk averse**.
- Over the last 50 years in the U.S., real stock returns have averaged 9% with a std of about 16%, while the real return on T-bills has been about 1%.

- Thus, the historical annual market Sharpe ratio has been about 0.5.
- The aggregate nondurable and services consumption growth had a std. of 1%.
- **This fact can only be reconciled with $\gamma = 50$.**
- But the empirical estimates are between 2 and 10.
- This is the **"equity premium puzzle"**: the historical Sharpe ratio is too large for reasonable risk aversion and consumption volatility estimates.

1.5 The risk free rate puzzle

- The standard CCAPM framework also gives rise to the risk free rate puzzle. Recall from Equation 12,

$$r^f_{t,t+1} = -\ln(\rho) + \gamma \mathbb{E}_t(\Delta \ln c_{t+1}) - \frac{\gamma^2}{2}\sigma^2_t(\Delta \ln c_{t+1}). \tag{26}$$

- Using $\gamma = 5$ the risk free rate should be around 5% to 6% per year.
- The actually observed rate is less than 1%.

1.6 How to resolve the equity premium and risk free puzzles?

- Perhaps investors are much more **risk averse** than we may have thought.
  - This indeed resolves the equity premium puzzle.
  - But higher risk aversion parameter implies higher risk free rate. So higher risk aversion reinforces the risk free puzzle.
- Perhaps the stock returns over the last 50 years are **good luck** rather than equilibrium compensation for risk $\longmapsto$ the equity premium may diminish in several decades.
- Perhaps something is **wrong with the utility function** specification.
  - The CCAPM assumes *time-additive preferences*, which impose tight restrictions on the relation between the equity premium and the risk free rate.
  - Epstein and Zin (1989) and Weil (1990) propose *recursive intertemporal utility* functions and Duffie and Epstein (1992a) and Duffie and Epstein (1992b) introduce a *stochastic differential utility*. $\longmapsto$ Recursive preferences match the data better.
  - Rietz (1988) brings the possibility of low probability states of *economic disaster* that is better able to explain the observed equity premium.

- Perhaps using **aggregate consumption data is wrong** (discussed earlier).
  - Consumption data is too smooth and low frequency $\longmapsto$ lower frequency of returns could be used, or $\longmapsto$ more volatile proxies for consumption growth are needed (see Savov, 2011, 'garbage').
- Perhaps there are **more sources of risk** $\longmapsto$ multifactor models (next).

1.7 Multifactor Models

- The poor empirical performance of the single factor (C)CAPM motivates a search for alternative asset pricing models.
- Either include **more factors** or go along with **conditional models** or **both**.
- Multiple factors have been inspired by:
  - The **Arbitrage Pricing Theory** — APT — Ross (1976).
  - The **Intertemporal CAPM** — ICAPM — Merton (1973).
  - The **Investment CAPM** — q-theory of investment — Zhang (2017): high investment stocks should earn lower expected returns than low in-

vestment stocks, and that stocks with high expected profitability should earn higher expected returns than stocks with low expected profitability.

- Distinguishing between the APT and ICAPM is often confusing.
- Cochrane (2001) argues that the biggest difference between them for empirical work is in the inspiration of factors:
  - The APT suggests a *statistical analysis* of the covariance matrix of returns to find factors that characterize common movements.
  - The ICAPM puts some *economic meaning* to the selected factors.
- For instance, Richard Roll and Ross (1980) propose *factor analysis* of the covariance matrix of returns, and Connor and Korajczyk (1988) propose an *asymptotic principle components*.

1.7.1 The Fama-French Model

- The three-factor Fama-French model is attractive from an empirical perspective.
- Fama and French (1993) [FF93] show that the cross-sectional variation in expected returns can be captured using the following factors:

1. the return on the market portfolio in excess of the risk free rate (**MKT**)
2. a zero net investment (spread) portfolio that is long in small firm stocks and short in large firm stocks (**SMB**)
3. a zero net investment (spread) portfolio that is long in high book-to-market stocks and short in low book-to-market stocks (**HML**)

- The factors as well as description about constructions are available at Kenneth French's web site (`http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`).
- Fama and French (1996) show that their model is able to explain many of the market anomalies known back then - excluding the momentum effect.
- SMB and HML are based on six size book-to-market sorted portfolios.
- The six portfolios are constructed at the end of each June as the intersections of 2 portfolios formed on market equity and 3 portfolios formed on the ratio of book equity to market equity.
  - The size breakpoint for year t is the median NYSE market equity at the end of June of year t.
  - BE/ME for June of year t is the book equity for the last fiscal year end in t-1 divided by ME for December of t-1. The BE/ME breakpoints are

the 30th and 70th NYSE percentiles.

  - SMB is the average return on the three small portfolios minus the average return on the three big portfolios.
  - HML is the average return on the two value portfolios minus the average return on the two growth portfolios.

- FF93 argue that their factors are state variables in an ICAPM sense.
- Liew and Vassalou (2000) make a good case for the FF assertion:
  They find that the FF factors forecast GDP growth.
- In the end, the Fama-French factors are empirically based, lack any theoretical underpinning.

1.7.2 Other Multifactor Models

The Carhart Four-Factor Model

- Carhart (1997) proposes a four-factor model to evaluate performance of equity mutual funds — **MKT, SMB, HML**, and **WML**
- **WML is a momentum factor**,
- **WML is computed as the return differential of past 'winners' and 'losers'**. The winner portfolio contains the 10% best stock performers based on past returns and the loser portfolio contains the 10% worst performers.
- Carhart shows that profitability of "hot hands" based trading strategies (documented by Hendricks, Patel, and Zeckhauser, 1993) disappears when investment payoffs are adjusted by WML.
- The profitability of "smart money" based trading strategies in mutual funds (documented by Zheng (2000)) also disappears in the presence of WML.

The Pastor and Stambaugh Liquidity Factor

- Pástor and Stambaugh (2003) propose adding a liquidity factor.

- So far we have **five major factors to explain equity returns**
  1. **MKT** - 'market factor'
  2. **SMB** - 'size factor'
  3. **HML** - 'value factor' or 'book-to-market factor'
  4. **WML** - 'momentum factor'
  5. **Liquidity** - 'liquidity factor'
- All are available at WRDS (`https://wrds-web.wharton.upenn.edu/wrds/`).
- They are widely used both in the industry and academic research (especially the first 4) to evaluate investment performance, compare new models, evaluate new factors, adjust for risk, and assess robustness of anomalies.
- In addition, **two factors are often used to explain bond returns**:
  1. **Term premium factor** – return differential between long- and short-term bonds – typically the 10-year and 1-year rates.
  2. **Default premium factor** - return differential between BBB- and AAA-rated bonds.

1.8 Do multifactor models really perform well?

- The sad news is that SMB, HML, WML, and the liquidity factors do not capture the size, book-to-market, past return, and liquidity effects in individual stock returns.
- Nor do they capture the many anomalies we will study in this class (see e.g. Avramov, Chordia, Jostova, and Philipov, 2013).

1.9 Do conditional models perform any better?

- Petkova and Zhang (2005) evaluated the conditional CAPM conditioning both betas and the risk premium on macro-level information.
- **The conditioning variables** (standard in the time-series predictability literature) are **the dividend yield (DIV), the default spread (DEF), the term spread (TERM), and the short-term Treasury bill rate (TB).**
- Petkova and Zhang (2005) conclude that **'value is riskier than growth'**—value stocks earn higher returns because they have higher systematic risk than growth stocks—providing a risk-based explanation for an asset pricing anomaly (more on this later).

- In particular, Petkova and Zhang (2005) estimate the risk premium in a time-series regression:

$$r_{mt+1} = \delta_0 + \delta_1\, DIV_t + \delta_2\, DEF_t + \delta_3\, TERM_t + \delta_4\, TB_t + e_{mt+1} \quad (27)$$

- **The estimate of the risk-premium is then:**

$$\widehat{\gamma}_t = \widehat{\delta}_0 + \widehat{\delta}_1\, DIV_t + \widehat{\delta}_2\, DEF_t + \widehat{\delta}_3\, TERM_t + \widehat{\delta}_4\, TB_t \quad (28)$$

- Conditional (time-varying) betas are estimated either through
  1. 24-, 36-, 48-, or 60-month **rolling regressions** or
  2. **conditional market regressions**:

$$r_{it+1} = \alpha_i + (b_{i0} + b_{i1}\, DIV_t + b_{i2}\, DEF_t + b_{i3}\, TERM_t + b_{i4}\, TB_t) r_{mt+1} + \epsilon_{it+1} \quad (29)$$

  and

$$\widehat{\beta}_t = \widehat{b}_{i1} + \widehat{b}_{i1}\, DIV_t + \widehat{b}_{i2}\, DEF_t + \widehat{b}_{i3}\, TERM_t + \widehat{b}_{i4}\, TB_t \quad (30)$$

- They then test the model using the predicted $\widehat{\gamma}$ and $\widehat{\beta}$.

- Alternatively, Avramov and Chordia (2006) examine pricing abilities of conditional and unconditional models, by **conditioning beta both on macro-level information** and **on firm-level characteristics**—beta is time-varying, changing with aggregate and firm-level conditions.
  - First, for each security, they estimate a time-series specification of the form

$$r_{it} = \alpha_i + \underbrace{\beta_i(\mathbf{z}_{t-1}, \mathbf{z}_{it-1})'}_{conditioning} f_t + \epsilon_{it} \tag{31}$$

    where $\mathbf{z}_t$ **is a macro-level conditioning information** and $\mathbf{z}_{it}$ **denotes a firm-level characteristic**, e.g. size, book-to-market.
  - **The risk-adjusted return is obtained as the intercept and residual from (31)**

$$r^*_{i,t} = \alpha_i + \epsilon_{it} \tag{32}$$

  - They then consider **cross-sectional regressions of risk-adjusted returns** on stock-level size, book-to-market, past return variables, and turnover.
  - If the pricing model does a good job in pricing individual stocks then **the slope coefficients in the cross-sectional regressions should**

be statistically indistinguishable from zero $\longmapsto$ the slope coefficients are, however, highly significant.

- That is, firm-level size, book-to-market, past return, and turnover are still important even when stock returns are adjusted by the SMB, HML, WML, and liquidity factors.
- We will revisit the Avramov and Chordia (2006) methodology in the section dealing with asset-pricing test methodologies.

- What about more factors?
  - Fama and French (2015) [JFE], *"A Five-Factor Asset Pricing Model"*—performs better than the Fama and French (1993) 3-factor model. Additional factors: profitability=RMW and investment=CMA.
  - Fama and French (2016) [RFS], *"Dissecting Anomalies with a Five-Factor Model"*—helps explain low-beta, share issues, and IVOL anomaly, but not others.
  - Hou, Xue, and Zhang (2015) [RFS], *"Digesting Anomalies: An Investment Approach"*—propose an alternative 5-factor model and looks at 80 anomalies—half are attenuated.
  - Harvey, Liu, and Zhu (2016) [RFS], *"... and the Cross-Section of Expected Returns"*—too much data-mining, "a new factor needs to clear

a much higher hurdle, with a t-statistic greater than 3.0."

1.10 Quick summary

- So the empirical evidence on model pricing abilities is disappointing
  $\longmapsto$ whether you take single- or multi-factor models
  $\longmapsto$ whether you take conditional or unconditional models.

- It is still important to develop asset pricing theories and test methodologies.

References for Asset Pricing Models

1 William F. Sharpe. "Capital Asset Prices: A Theory of Market Equilibrium". In: *Journal of Finance* 19 (1964), pp. 425–442.

2 John Lintner. "The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets". In: *Review of Economics and Statistics* 47 (1965), pp. 13–37.

3 Ray Ball and Philip Brown. "An Empirical Evaluation of Accounting Income Numbers". In: *Journal of Accounting Research* 6.2 (Sept. 1968), pp. 159–178.

4 Robert C. Merton. "An Intertemporal Capital Asset Pricing Model". In: *Econometrica* 41.5 (Sept. 1973), pp. 867–887.

5 Stephen A. Ross. "The Arbitrage Theory of Capital Asset Pricing". In: *Journal of Economic Theory* 13 (1976), pp. 341–360.

6 Mark Rubinstein. "The Strong Case for the Generalized Logarithmic Utility Model as the Premier Model of Financial Markets". In: *Journal of Finance* 31.2 (May 1976), pp. 551–571.

7 S. Basu. "Investment Performance of Common Stocks in Relation to Their Price-Earnings Ratios: A Test of the Efficient Market Hypothesis". In: *Journal of Finance* 32.3 (June 1977), pp. 663–682.

8 R. Roll. "A Critique of the Asset Pricing Theory's Tests: on Past and Potential Testability of Theory". In: *Journal of Financial Economics* 4 (1977), pp. 129–176.

9 Robert E. Lucas. "Asset Prices in an Exchange Economy". In: *Econometrica* 46.6 (Nov. 1978), pp. 1429–1445.

10 Douglas T. Breeden. "An Intertemporal Asset Pricing Model with Stochastic Consumption and Investment Opportunities". In: *Journal of Financial Economics* 7 (1979), pp. 265–296.

11 Richard Roll and Stephen A. Ross. "An Empirical Investigation of the Arbitrage Pricing Theory". In: *Journal of Finance* 35.5 (Dec. 1980), pp. 1073–1103.

12 Rolf W. Banz. "The Relationship between Return and Market Value of Common Stocks". In: *Journal of Financial Economics* 9 (1981), pp. 3–18.

13 Lars Peter Hansen and Kenneth J. Singleton. "Generalized Instrumental Variables Estimation of Nonlinear Rational Expectations Models". In: *Econometrica* 50.5 (Sept. 1982), pp. 1269–1286.

14 Lars Peter Hansen and Kenneth J. Singleton. "Stochastic Consumption, Risk Aversion, and the Temporal Behavior of Asset Returns". In: *Journal of Political Economy* 91.2 (1983), pp. 249–265.

15 Werner F. M. De Bondt and Richard H. Thaler. "Does the Stock Market Overreact?" In: *Journal of Finance* 40.3 (July 1985), pp. 793–805.

16 Rajnish Mehra and Edward C. Prescott. "The Equity Premium: A Puzzle". In: *Journal of Monetary Economics* 15.2 (Mar. 1985), pp. 145–161.

17 Nai-Fu Chen, Richard Roll, and Stephen A. Ross. "Economic Forces and the Stock Market". In: *Journal of Business* 59.3 (July 1986), pp. 383–403.

18 N. Gregory Mankiw and Matthew D. Shapiro. "Do We Reject Too Often? Small Sample Properties of Tests of Rational Expectations Models". In: *Economic Letters* 20 (1986), pp. 139–145.

19 Werner F. M. De Bondt and Richard H. Thaler. "Further Evidence on Investor Overreaction and Stock Market Seasonality". In: *Journal of Finance* 42.3 (July 1987), pp. 557–581.

20 Gregory Connor and Robert A. Korajczyk. "Risk and Return in an Equilibrium APT". In: *Journal of Financial Economics* 21 (1988), pp. 255–289.

21 Sanford J. Grossman and Robert J. Shiller. "The Determinants of the Variability of Stock Market Prices". In: *American Economic Review* 71.2 (1988), pp. 222–227.

22 Thomas A. Rietz. "The Equity Premium Puzzle: A Solution". In: *Journal of Monetary Economics* 22.1 (July 1988), pp. 117–131.

23 Simon Wheatley. "Some tests of international equity integration". In: *Journal of Financial Economics* 21.2 (Sept. 1988), pp. 177–212.

24 Douglas T. Breeden, Michael R. Gibbons, and Robert H. Litzenberger. "Empirical Test of the Consumption-Oriented CAPM". In: *Journal of Finance* 44.2 (June 1989), pp. 231–262.

25 Larry G. Epstein and Stanley E. Zin. "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework". In: *Econometrica* 57.4 (July 1989), pp. 937–969.

26 Narasimhan Jegadeesh. "Evidence of Predictable Behavior of Security Returns". In: *Journal of Finance* 45.3 (July 1990), pp. 881–898.

27 B. Lehmann. "Fads, martingales, and market efficiency". In: *Quarterly Journal of Economics* 60 (1990), pp. 1–28.

28 Andrew W. Lo and A. Craig MacKinlay. "When are Contrarian Profits Due to Stock Market Overreaction?" In: *Review of Financial Studies* 3.2 (1990), pp. 175–205.

29 Philippe Weil. "Non-Expected Utility in Macroeconomics". In: *Quarterly Journal of Economics* 105.1 (Feb. 1990), pp. 29–42.

30 Darrell Duffie and Larry G. Epstein. "Asset Pricing with Stochastic Differential Utility". In: *Review of Financial Studies* 5.3 (1992), pp. 411–436.

31 Darrell Duffie and Larry G. Epstein. "Stochastic Differential Utility". In: *Econometrica* 60.2 (Mar. 1992), pp. 353–394.

32 Eugene F. Fama and Kenneth R. French. "The Cross-Section of Expected Stock Returns". In: *Journal of Finance* 47.2 (June 1992), pp. 427–465.

33 Eugene F. Fama and Kenneth R. French. "Common Risk Factors in the Returns on Stocks and Bonds". In: *Journal of Financial Economics* 33.1 (Feb. 1993), pp. 3–56.

34 Darryll Hendricks, Jayendu Patel, and Richard Zeckhauser. "Hot Hands in Mutual Funds: Short-Run Persistence of Relative Performance, 1974-1988". In: *Journal of Finance* 48.1 (Mar. 1993), pp. 93–130.

35 Narasimhan Jegadeesh and Sheridan Titman. "Returns to buying winners and selling losers: implications for stock market efficiency". In: *Journal of Finance* 48.1 (Mar. 1993), pp. 65–91.

36 John H. Cochrane. "A Cross-Sectional Test of An Investment-Based Asset Pricing Model". In: *Journal of Political Economy* 104.3 (June 1996), pp. 572–621.

37 Eugene F. Fama and Kenneth R. French. "Multifactor Explanations of Asset Pricing Anomalies". In: *Journal of Finance* 51.1 (Mar. 1996), pp. 55–84.

38 Richard G. Sloan. "Do Stock Prices Fully Reflect Information in Accruals and Cash Flows About Future Earnings?" In: *The Accounting Review* 71.3 (1996), pp. 289–316.

39 Mark M. Carhart. "On Persistence in Mutual Fund Performance". In: *Journal of Finance* 52.1 (Mar. 1997), pp. 57–82.

40 Ilia D. Dichev. "Is the Risk of Bankruptcy a Systematic Risk?" In: *Journal of Finance* 53.3 (1998), pp. 1131–1147.

41 K. Geert Rouwenhorst. "International Momentum Strategies". In: *Journal of Finance* 53.1 (1998), pp. 267–284.

42 Jimmy Liew and Maria Vassalou. "Can book-to market, size and momentum be risk factors that predict economic growth?" In: *Journal of Financial Economics* 57.2 (Aug. 2000), pp. 221–245.

43 Lu Zheng. "Is Money Smart? A Study of Mutual Fund Investors' Fund Selection Ability". In: *Journal of Finance* 54.3 (June 2000), pp. 01–933.

44 John H. Cochrane. *Asset Pricing*. Princeton, New Jersey: Princeton University Press, 2001.

45 Ilia D. Dichev and Joseph D. Piotroski. "The Long-Run Stock Returns Following Bond Rating Changes". In: *Journal of Finance* 56.1 (2001), pp. 55–84.

46 William Fung and David A. Hsieh. "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers". In: *Review of Financial Studies* 14.2 (Summer 2001), pp. 313–341.

47 Martin Lettau and Sydney L. Ludvigson. "Consumption, Aggregate Wealth, and Expected Stock Returns". In: *Journal of Finance* 56.3 (June 2001), pp. 815–849.

48 Martin Lettau and Sydney L. Ludvigson. "Resurrecting the (C)CAPM: A Cross-Sectional Test When Risk Premia Are Time-Varying". In: *Journal of Political Economy* 109.6 (Dec. 2001), pp. 1238–1287.

49 Karl B. Diether, Christopher J. Malloy, and Anna Scherbina. "Difference of Opinion and the Cross Section of Stock Returns". In: *Journal of Finance* 57.5 (2002), pp. 2113–2141.

50 Luboš Pástor and Robert F. Stambaugh. "Liquidity Risk and Expected Stock Returns". In: *Journal of Political Economy* 111.3 (2003), pp. 642–685.

51 Yaron Amir and Ravi Bansal. "Risks for the Long Run: A Potential Resolution of Asset Pricing Puzzles". In: *Journal of Finance* 59.4 (Aug. 2004), pp. 1481–1509.

52 William Fung and David A. Hsieh. "Hedge Fund Benchmarks: A Risk-Based Approach". In: *Financial Analysts Journal* 60.5 (Sept. 2004), pp. 65–80.

53 Sheridan Titman, K. C. John Wei, and Feixue Xie. "Capital Investments and Stock Returns". In: *Journal of Financial and Quantitative Analysis* 39.04 (Dec. 2004), pp. 677–700.

54 Ravi Bansal, Robert F. Dittmar, and Christian T. Lundblad. "Consumption, Dividends, and the Cross-Section of Equity Returns". In: *Journal of Finance* 60.4 (Aug. 2005), pp. 1639–1672.

55 Gergana Jostova and Alexander Philipov. "Bayesian Analysis of Stochastic Betas". In: *Journal of Financial and Quantitative Analysis* 40.4 (2005), pp. 747–778.

56 Jonathan A. Parker and Christian Julliard. "Consumption Risk and the Cross Section of Expected Returns". In: *Journal of Political Economy* 113.1 (Feb. 2005), pp. 185–222.

57 Ralitsa Petkova and Lu Zhang. "Is Value Riskier than Growth?" In: *Journal of Financial Economics* 78.1 (Oct. 2005).

58 Andrew Ang, Robert J. Hodrick, Yuhang Xing, and Xiaoyan Zhang. "The Cross-Section of Volatility and Expected Returns". In: *Journal of Finance* 61.1 (Feb. 2006), pp. 259–299.

59 Doron Avramov and Tarun Chordia. "Asset Pricing Models and Financial Market Anomalies". In: *Review of Financial Studies* 19.3 (2006), pp. 1001–1040.

60 Doron Avramov, Tarun Chordia, and Amit Goyal. "The Impact of Trades on Daily Volatility". In: *Review of Financial Studies* 19.4 (Jan. 2006), pp. 1241–1277.

61 Malcolm Baker and Jeffrey Wurgler. "Investor Sentiment and the Cross-Section of Stock Returns". In: *Journal of Finance* 61.4 (Aug. 2006), pp. 1645–1680.

62 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Momentum and Credit Rating". In: *Journal of Finance* 62.5 (Oct. 2007), pp. 2503–2520.

63 John Y. Campbell, Jens Hilscher, and Jan Szilagyi. "In Search of Distress Risk". In: *Journal of Finance* 63.6 (Dec. 2008), pp. 2899–2939.

64 Michael J. Cooper, Huseyin Gulen, and Michael J. Schill. "Asset Growth and the Cross-Section of Stock Returns". In: *Journal of Finance* 63.4 (Aug. 2008), pp. 1609–1651.

65 Eugene F. Fama and Kenneth R. French. "Dissecting Anomalies". In: *Journal of Finance* 63.4 (Aug. 2008), pp. 1653–1678.

66 Lars Peter Hansen, John Heaton, and Nan Li. "Consumption Strikes Back? Measuring Long-Run Risk". In: *Journal of Political Economy* 116.2 (Apr. 2008), pp. 260–302.

67 Andrew Ang, Robert J. Hodrick, Yuhang Xing, and Xiaoyan Zhang. "High Idiosyncratic Volatility and Low Returns: International and Further U.S. Evidence". In: *Journal of Financial Economics* 91.1 (Jan. 2009), pp. 1–23.

68 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Credit Ratings and The Cross-Section of Stock Returns". In: *Journal of Financial Markets* 12 (2009), pp. 469–499.

69 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Dispersion in Analysts' Earnings Forecasts and Credit Rating". In: *Journal of Financial Economics* 91.1 (Jan. 2009), pp. 83–101.

70 Joao F. Gomes and Lukas Schmid. "Equilibrium Credit Spreads and the Macroeconomy". In: *SSRN eLibrary, http://ssrn.com/paper=1363* (2010).

71 Alexi Savov. "Asset Pricing with Garbage". In: *Journal of Finance* 66.1 (Feb. 2011), pp. 177–201.

72 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "The World Price of Credit Risk". In: *Review of Asset Pricing Studies* 2.2 (Dec. 2012), pp. 112–152.

73 Clifford S. Asness, Tobias J. Moskowitz, and Lasse H. Pedersen. "Value and Momentum Everywhere". In: *Journal of Finance* 68.3 (June 2013), pp. 929–985.

74 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Anomalies and Financial Distress". In: *Journal of Financial Economics* 108.2 (May 2013), pp. 139–159.

75 Gergana Jostova, Stanislava Nikolova, Alexander Philipov, and Christof Stahel. "Momentum in corporate bond returns". In: *Review of Financial Studies* 26.7 (May 2013), pp. 1649–1693.

76 Eugene F. Fama and Kenneth R. French. "A Five-Factor Asset Pricing Model". In: *Journal of Financial Economics* 116.1 (Sept. 2015), pp. 1–22.

77 Kewei Hou, Chen Xue, and Lu Zhang. "Digesting Anomalies: An Investment Approach". In: *Review of Financial Studies* 28.3 (2015), pp. 650–705.

78 Eugene F. Fama and Kenneth R. French. "Dissecting Anomalies with a Five-Factor Model". In: *Review of Financial Studies* 29.1 (Sept. 2016), pp. 69–103.

79 Campbell R. Harvey, Yan Liu, and Heqing Zhu. "... and the Cross-Section of Expected Returns". In: *Review of Financial Studies* 29.1 (2016), pp. 5–68.

80 R. David McLean and Jeffrey Pontiff. "Does Academic Research Destroy Stock Return Predictability?" In: *The Journal of Finance* 71.1 (Feb. 2016), pp. 5–32.

81 Lu Zhang. "The Investment CAPM". In: *European Financial Management* 23.4 (2017), pp. 545–603.

82 Kewei Hou, Chen Xue, and Lu Zhang. "Replicating anomalies". In: *Review of Financial Studies* 33.5 (May 2020), pp. 2019–2133.

83 Mark Grinblatt, Gergana Jostova, and Alexander Philipov. "Analyst Bias and Anomalies". In: *Critical Finance Review* forthcoming (2025). eprint: https://cfr.ivo-welch.info/forthcoming/papers/grinblatt2024analyst.pdf.